SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS: **Total Marks: 30**

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.

Questions:

1. Identify appropriate **HTML5 semantic elements** missing in the structure.
2. Modify the structure to make it **standards-compliant**.
3. Interpret the **HTTP request generated** during form submission.
4. Analyze the **HTTP response cycle** from server to browser.
5. Suggest improvements for **usability and accessibility**.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
  <head>
    <title>Company</title>
  </head>
  <body>
    <div>
      <h1>Welcome
      <p>About us
    </div>
  </body>
</html>
```

Questions:

1. Identify **improper nesting and missing closing tags**.
2. Analyze how different browsers may interpret this structure.
3. Identify **missing semantic elements** (e.g., header, section).
4. Provide a **corrected W3C-compliant HTML structure**.
5. Suggest techniques to ensure **cross-browser compatibility**.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="button">Login</button>
</form>
```

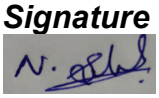
Questions:

1. Identify issues in the **form configuration**.
2. Analyze why the **HTTP request is not triggered**.
3. Interpret the role of **GET method in this scenario**.
4. Propose a **corrected implementation**.
5. Suggest improvements for **secure data transmission**.

Code of Conduct:

I Shagithiyan N 192324259 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



QA Scenario: Web Development Standards Implementation

A start-up is developing a modern web application and faces issues related to HTML elements, HTTP methods, XML syntax and web communication protocols. Each issue is analysed below, the correct option is selected, and the reasoning is given.

Question 1: HTML tag for embedding a promotional video (Snippet 1)

Correct answer: Option B – `<video src="video.mp4" controls></video>`

Justification:

- <video> is the standard element introduced in HTML5 for embedding video natively in a web page, without third-party plug-ins such as Flash.
- The src attribute gives the file location, and the controls attribute shows the play/pause, volume and full-screen buttons to the user.
- Options A (<media>), C (<movie>) and D (<vid>) are not valid HTML elements. Browsers ignore them and the video would not be displayed.

Improved example with a fallback and CSS for responsiveness:

```
<video width="640" height="360" controls poster="thumb.jpg">
  <source src="video.mp4" type="video/mp4">
  Your browser does not support the video tag.
</video>
video { max-width: 100%; height: auto; }
```

Question 2: HTTP method for form submission (Snippet 2)

Correct answer: Option B – POST

Justification:

- POST sends the form data in the body of the HTTP request, so values such as passwords and email IDs are not shown in the URL, browser history or server logs. GET appends data to the URL as a query string, which is visible and unsafe for registration data.
- POST has no practical limit on data size, whereas GET is limited by URL length.
- POST is used to create a new resource (a new user account) on the server. It is not idempotent, which suits registration.
- PUT is used to replace or update an existing resource and DELETE is used to remove a resource. Neither is supported by an HTML form's method attribute, and neither fits user registration.
- Note: POST alone does not encrypt data. For real security it must be combined with HTTPS.

```
<form action="/register" method="post">
  <input type="text" name="username" required>
  <input type="password" name="password" required>
  <button type="submit">Register</button>
</form>
```

Question 3: Semantic element for navigation (Snippet 3)

Correct answer: Option B – <nav>

Justification:

- <nav> is the HTML5 semantic element meant specifically for a block of major navigation links such as a main menu.

- It gives meaning to the page structure, which helps screen readers and assistive technologies jump to the menu (accessibility), helps search engines understand the layout (SEO), and makes the code easier to read and maintain.
- is a non-semantic generic container, <section> groups thematic content, and <header> holds introductory content or a heading area. None of them identifies a navigation menu.

```
<nav>
  <ul>
    <li><a href="index.html">Home</a></li>
    <li><a href="about.html">About</a></li>
    <li><a href="contact.html">Contact</a></li>
  </ul>
</nav>
```

Question 4: Correct XML syntax (Snippet 4)

Correct answer: Option B – `<note><to>User</to><from>Admin</from></note>`

Option B is well-formed: it has a single root element, every start tag has a matching end tag, and the elements are properly nested. Errors in the other options:

Option	Code	Error
A	<note><to>User</to><from>Admin</from>	The root element <note> is never closed (missing </note>).
C	<note><to>User</from></note>	Tags are mismatched: <to> is closed with </from>, so nesting is improper and the <from> element does not exist properly.
D	<note to="User"></note	The closing tag is incomplete (missing '>'), so the markup is malformed.

XML is strict: tags are case-sensitive, must be closed and properly nested, attribute values must be quoted, and a document must have exactly one root element.

Question 5: Protocol for web communication (Snippet 5)

Correct answer: Option B – HTTP

Justification:

- HTTP (HyperText Transfer Protocol) is the application-layer protocol used by browsers and web servers to request and deliver web pages, images, videos, CSS and other resources. It follows a request-response model (GET, POST, etc.) over TCP, usually on port 80. Its secure version, HTTPS (port 443), adds TLS encryption and should be used for the registration form.
- FTP is used for file transfer, SMTP is used for sending email, and SNMP is used for monitoring and managing network devices. None of these is used to deliver web content to a browser.